

Assignment - 2

①

$$m \ddot{x} + k \dot{x} |\dot{x}| = u$$

(a) state space model.

$$\text{Let } x_1 = x, \quad x_2 = \dot{x}$$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\frac{k}{m} x_2 |x_2| + \frac{1}{m} u \end{cases}$$

(b) Conventional Sliding Mode Control

$$s = c x_1 + x_2, \quad c > 0$$

$$\dot{s} = c \dot{x}_1 + \dot{x}_2$$

$$= c x_2 - \frac{k}{m} x_2 |x_2| + \frac{u}{m}$$

Proposed control law

$$u = -m c x_2 - K x_2 \operatorname{sign}(s)$$

$$\dot{s} = -\frac{K}{m} x_2 |x_2| - \frac{K}{m} \operatorname{sign}(s)$$

Reaching condition

$$s \dot{s} = -\frac{K}{m} x_2 |x_2| s - \frac{K}{m} |s|$$

$$\therefore \left| -\frac{K}{m} x_2 |x_2| s \right| \leq \frac{K}{m} \dot{x}_2^2 |s|$$

$$\therefore \dot{S} \leq \frac{k_2}{m} \dot{x}_2^2 |S| - \frac{K}{m} |S|$$

$$= \frac{1}{m} (k_2 \dot{x}_2^2 - K) |S|$$

$$\eta\text{-reachability} \Rightarrow \dot{S} \leq -\eta |S|, \text{ with } K \geq k_2 \dot{x}_2^2 + m\eta$$

Control law:

$$\begin{cases} u = -m c x_2 - (k_2 \dot{x}_2^2 + m\eta) \operatorname{sign}(S), \eta > 0 \end{cases}$$

(d.)

$$S = c x_1 + x_2, \quad c > 0$$

$$\dot{S} = c \dot{x}_1 + \dot{x}_2$$

$$= c x_2 - \frac{k}{m} x_2 |x_2| + \frac{u}{m} + \dot{x}_2 = \dot{x}_2$$

$$V = \frac{1}{2} S^2 > 0$$

$$\dot{V} = S \dot{S} = \frac{k}{m} x_2 |x_2| S - \frac{K}{m} |S|$$

$$(\text{using hint}) \left| \frac{k}{m} x_2 |x_2| \cdot S \right| \leq \frac{k}{m} |S| \dot{x}_2^2$$

$$\dot{V} \leq \left(\frac{k \dot{x}_2^2}{m} - \frac{K}{m} \right) |S|$$

Hence, the sliding variable (S) reaches zero in finite time.

$$s \geq 0 \Rightarrow x_2 = -cx_1$$

$$\dot{x}_1 = x_2 = -cx_1$$

$$x_1(t) = x_1(0) e^{-ct}, \quad c > 0$$

$$x_1(t) \rightarrow 0 \text{ as } t \rightarrow \infty$$

and

$$x_2 = -cx_1$$

$$= -cx_1(t) e^{-ct}, \quad c > 0$$

$$x_2(t) \rightarrow 0 \text{ as } t \rightarrow \infty$$

so, states x_1, x_2 reaches zero asymptotically

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

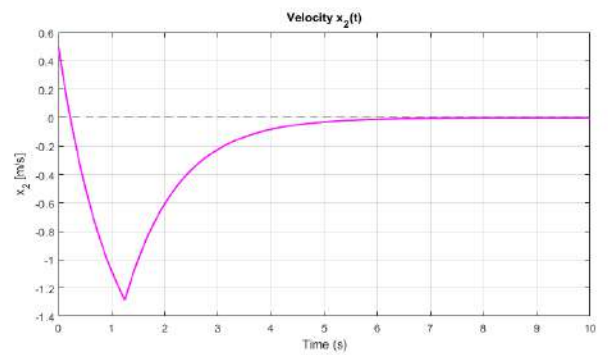
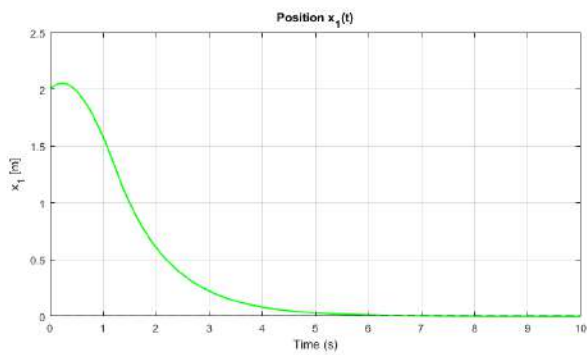
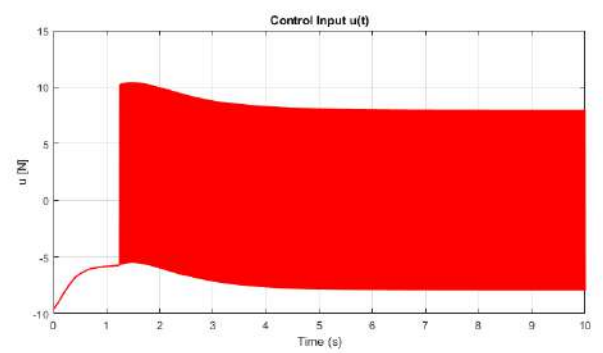
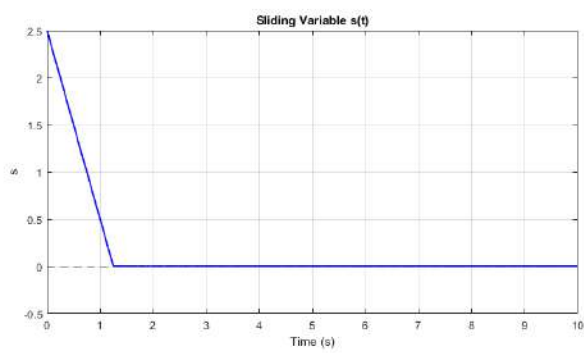
$$Q = I$$

$$Q = I$$

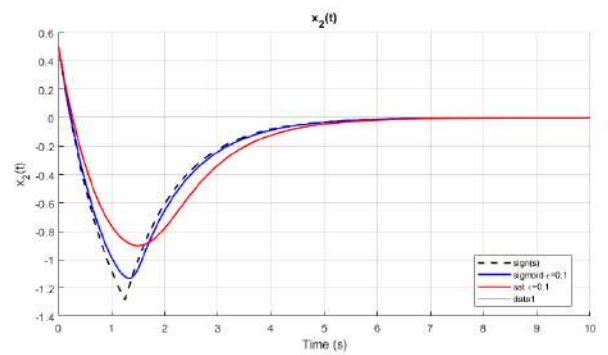
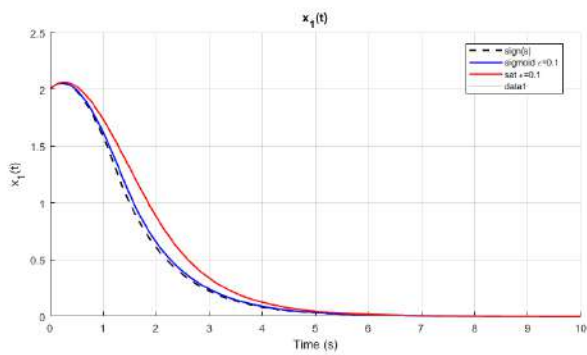
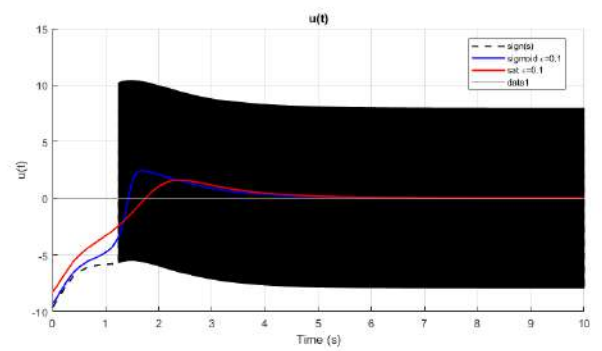
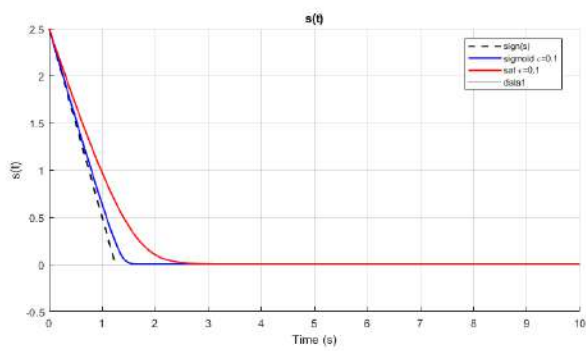
$$Q = I$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = u, \quad \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = v$$

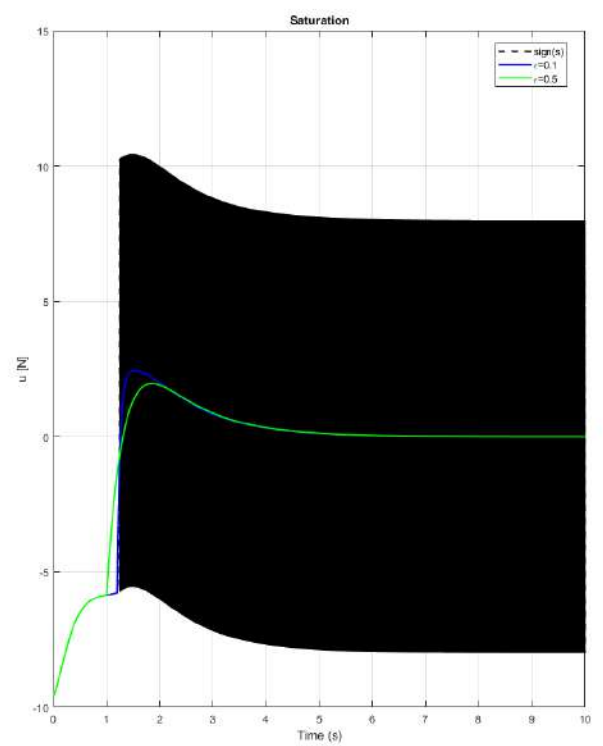
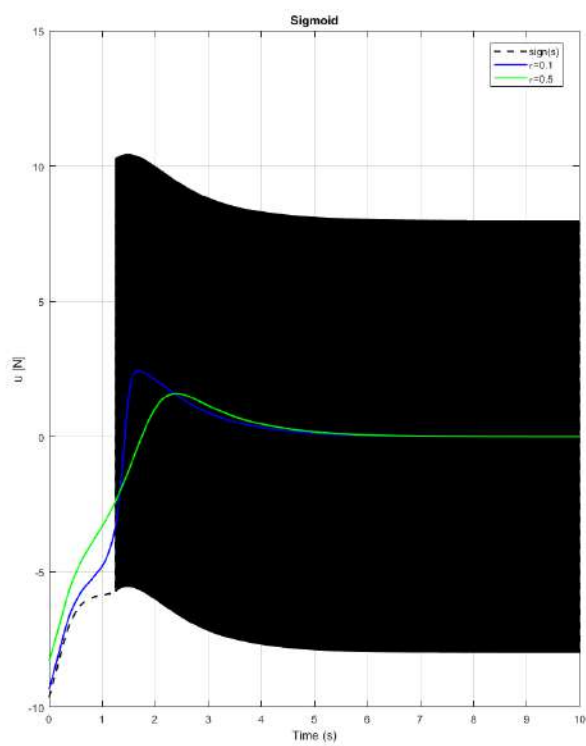
Solution 1(c)



Solution 2.



Solution 2.



3.

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 3 & 5 \end{bmatrix} x + \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} (u+d)$$

$$x \in \mathbb{R}^3, u \in \mathbb{R}^2, d \in \mathbb{R}^2, d = \begin{bmatrix} 2 \sin(t) + \cos(t) \\ 1 + \cos(t) \end{bmatrix}$$

(A) Equilibrium point for $d=0$

at equilibrium $\dot{x} = 0$

$$Ax + Bu = 0$$

for $u=0$

$$Ax = 0$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 3 & 5 \end{bmatrix} x = 0$$

$$x_2 = 0$$

$$x_3 = 0$$

$$x_1 + 3x_2 + 5x_3 = 0 \Rightarrow x_1 = 0$$

$$\left\{ x_{eq} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, u = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right.$$

Continuous State Feedback Controller

$$u = -Kx.$$

check controllability

$$C = [B \quad AB \quad A^2B]$$

$$= \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 0 & 1 & 2 \\ 1 & 2 & 10 & 11 \end{bmatrix}$$

Rank = 3 $\rightarrow$ controllable

$$\begin{cases} u = -Kx. \end{cases}$$

Sliding Mode Controller

$$s = Cx, \quad C \in \mathbb{R}^{2 \times 3}$$

$$\dot{s} = C\dot{x} = C(Ax + B(u+d))$$

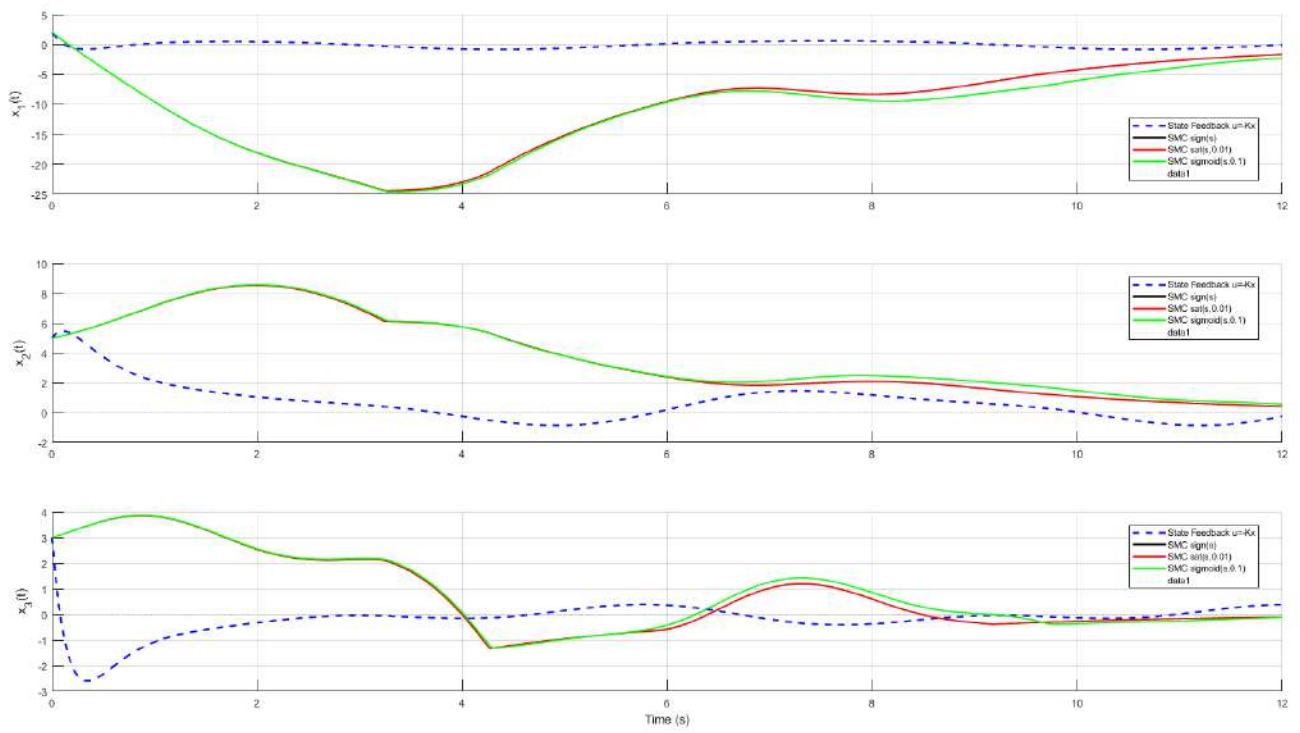
$$\dot{s} = CAx + CB(u+d)$$

$$\dot{s} = 0 \Rightarrow u = -(CB)^{-1} CAx.$$

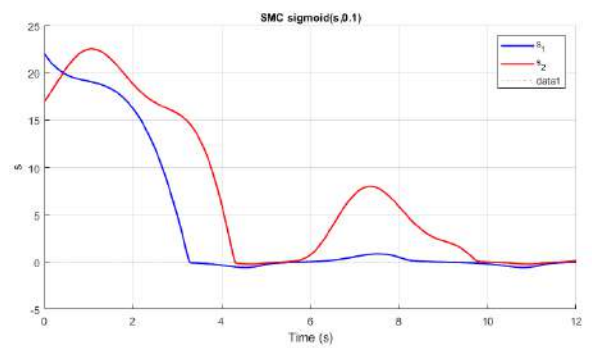
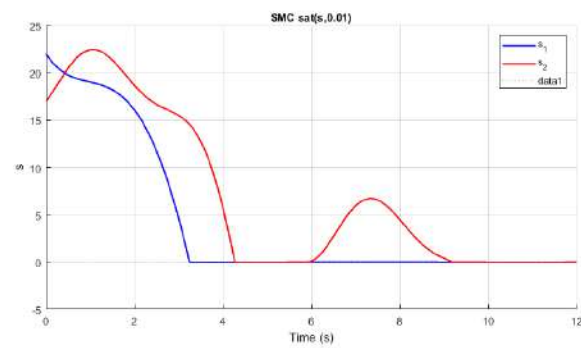
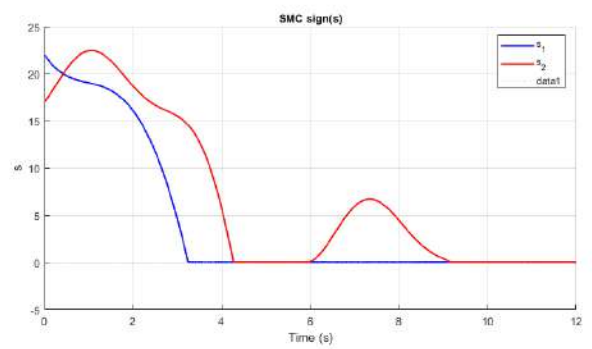
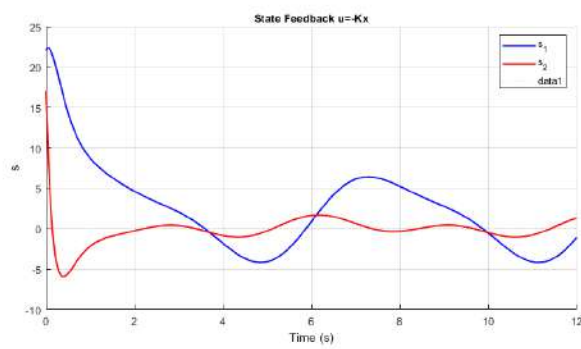
for $-K \text{ sign}(s)$

$$u = -(CB)^{-1} CAx - K \text{ sign}(s)$$

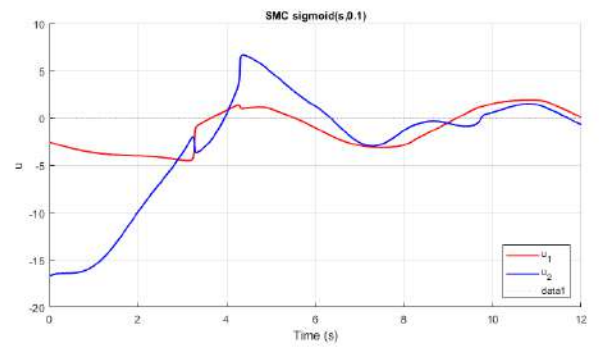
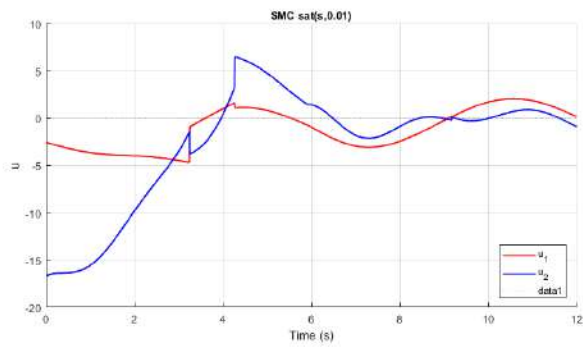
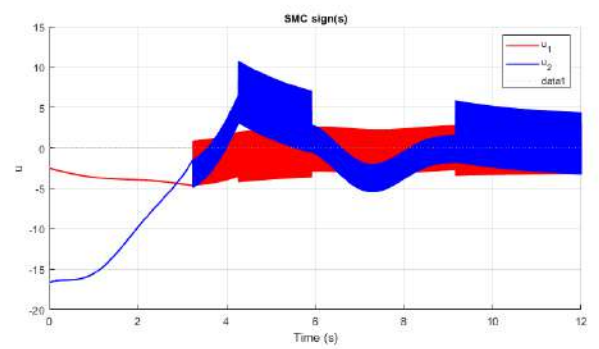
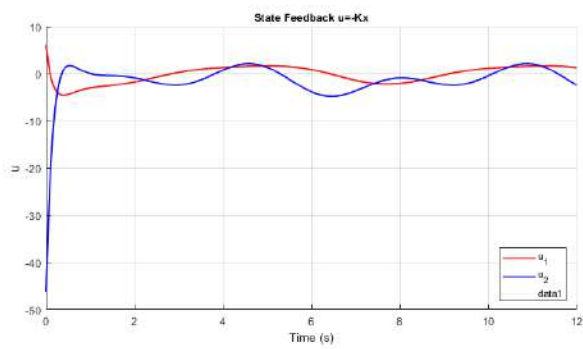
Solution 3. States



Solution 3. Sliding surfaces



Solution 3. Control inputs



④

NTSM.

$$s = x_1 + \frac{1}{\beta} \text{sign}(x_2) |x_2|^{\frac{p}{q}}$$

$$\dot{s} = \dot{x}_1 + \frac{1}{\beta} \frac{p}{q} |x_2|^{\frac{p}{q}-1} \dot{x}_2$$

$$= x_2 + \frac{1}{\beta} \frac{p}{q} |x_2|^{\frac{p}{q}-1} [f(x) + g(x)(u+d)]$$

set $\dot{s} = -k \text{sign}(s)$

$$\begin{cases} u = -\beta \frac{q}{p} \times \frac{1}{g} |x_2|^{2-\frac{p}{q}} \text{sign}(x_2) - \frac{f(x)}{g} - \beta \frac{q}{p} \frac{1}{g} |x_2|^{1-\frac{p}{q}} \text{sign}(s) \end{cases}$$

Let $V = \frac{1}{2} s^2$

$$\dot{V} = s \dot{s} = s(-k \text{sign}(s) + d)$$

$$= -k |s| + d s \leq -(k - g d_{\max}) |s|$$

with $k > g d_{\max} + \eta$.

$$\dot{V} \leq -\eta |s| = -\eta \sqrt{2V} \rightarrow \text{finite time stable}$$

convergence time.

After $\delta = 0$

$$\dot{x}_2 = -\frac{1}{p} \beta |x_2|^{2-\frac{p}{q}} \text{sign}(x_2).$$

let $\beta = 1$, $p = 5$, $q = 3$.

$$\dot{x}_2 = -\frac{3}{5} |x_2|^{\frac{1}{3}} \text{sign}(x_2)$$

$$\left. \frac{3}{2} x_2^{\frac{2}{3}} \right|_{x_2}^0 = \frac{3}{5} T$$

$$T = \frac{5}{2} |x_2(t)|^{\frac{2}{3}}.$$

for $t_2 \rightarrow |s(t)| \leq |s(0)| - \eta t$.

$$t = \frac{|s(0)|}{\eta}$$

$$\left\{ \begin{array}{l} T_{\text{total}} = \frac{|s(0)|}{\eta} + \frac{5}{2} |x_2(t)|^{\frac{2}{3}} \\ , \end{array} \right.$$

```
% Q. 1
clear; clc; close all;

m = 4;
c = 1;
eta = 2;
k_func = @(t) 1.5 + 0.4*sin(2*t);

ode_smc = @(t, x) dynamics(t, x, m, c, eta, k_func);

x0 = [2; 0.5];
tspan = [0 10];
opts = odeset('MaxStep', 0.001, 'RelTol', 1e-6);
[t, X] = ode45(ode_smc, tspan, x0, opts);

s_arr = c*X(:,1) + X(:,2);
u_arr = zeros(length(t),1);
for i = 1:length(t)
    k = k_func(t(i));
    x2 = X(i,2);
    s = s_arr(i);
    u_arr(i) = k*x2*abs(x2) - m*c*x2 - m*eta*sign(s);
end

figure('Name','Q1 - SMC Underwater Vehicle','Position',[100 100 1000 700]);

subplot(2,2,1)
plot(t, s_arr, 'b', 'LineWidth', 1.5); hold on
yline(0, 'k--', 'LineWidth', 0.8);
xlabel('Time (s)'); ylabel('s'); title('Sliding Variable s(t)');
grid on;

subplot(2,2,2)
plot(t, u_arr, 'r', 'LineWidth', 1.2);
xlabel('Time (s)'); ylabel('u [N]'); title('Control Input u(t)');
grid on;

subplot(2,2,3)
plot(t, X(:,1), 'g', 'LineWidth', 1.5); hold on
yline(0, 'k--', 'LineWidth', 0.8);
xlabel('Time (s)'); ylabel('x_1 [m]'); title('Position x_1(t)');
grid on;

subplot(2,2,4)
plot(t, X(:,2), 'm', 'LineWidth', 1.5); hold on
yline(0, 'k--', 'LineWidth', 0.8);
xlabel('Time (s)'); ylabel('x_2 [m/s]'); title('Velocity x_2(t)');
grid on;

function dxdt = dynamics(t, x, m, c, eta, k_func)
    x1 = x(1); x2 = x(2);
```

```
k = k_func(t);  
s = c*x1 + x2;  
u = k*x2*abs(x2) - m*c*x2 - m*eta*sign(s);  
dx1 = x2;  
dx2 = (u - k*x2*abs(x2)) / m;  
dxdt = [dx1; dx2];  
end
```



```
clear; clc; close all;

A = [0 1 0;
      0 0 1;
      1 3 5];

B = [2 1;
      1 0;
      1 2];

d_func = @(t) [2*sin(t)+cos(t); 1+cos(2*t)];

% u = -K*x
desired_poles = [-3, -4, -5];
K_sf = place(A, B, desired_poles);
eigs_cl = eig(A - B*K_sf);

% SMC
C = [1 4 0; 0 1 4];
CB = C * B;
CB_inv = inv(CB);
K_eq = CB_inv * C * A;

% Disturbance bound -> choose eta
d_max = [3; 2];
eta = max(norm(CB_inv) * norm(C) * norm(d_max) + 1, 15);

x0 = [2; 5; 3];
tspan = [0 12];
opts = odeset('MaxStep', 0.001, 'RelTol', 1e-6);

% Run all four controllers
[t1, X1] = ode45(@(t,x) ode_q3(t,x,A,B,K_sf,K_eq,CB_inv,C,eta,d_func,'sf', 0), ↙
tspan, x0, opts);
[t2, X2] = ode45(@(t,x) ode_q3(t,x,A,B,K_sf,K_eq,CB_inv,C,eta,d_func,'sign', 0), ↙
tspan, x0, opts);
[t3, X3] = ode45(@(t,x) ode_q3(t,x,A,B,K_sf,K_eq,CB_inv,C,eta,d_func,'sat', 0.01), ↙
tspan, x0, opts);
[t4, X4] = ode45(@(t,x) ode_q3(t,x,A,B,K_sf,K_eq,CB_inv,C,eta,d_func,'sigmoid',0.1), ↙
tspan, x0, opts);

T = {t1,t2,t3,t4};
Xs = {X1,X2,X3,X4};
labels = {'State Feedback u=-Kx','SMC sign(s)','SMC sat(s,0.01)','SMC sigmoid(s,0.1)'};
cols = {'b','k','r','g'};
ls_arr = {'--','-', '-','-'};

% states
figure('Name','States','Position',[50 50 1100 850]);
state_names = {'x_1(t)','x_2(t)','x_3(t)'};
```

```

for si = 1:3
    subplot(3,1,si); hold on;
    for ci = 1:4
        plot(T{ci}, Xs{ci}(:,si), 'Color',cols{ci},'LineWidth',1.8,'LineStyle',ls_arr{
{ci},'DisplayName',labels{ci}};
    end
    yline(0,'Color',[0.6 0.6 0.6],'LineWidth',0.5,'LineStyle',':');
    ylabel(state_names{si},'FontSize',12); grid on;
    legend('Location','best','FontSize',8);
    if si==3; xlabel('Time (s)','FontSize',11); end
end

% Sliding Variables
figure('Name','Sliding Variables','Position',[50 50 1200 700]);
for ci = 1:4
    subplot(2,2,ci); hold on;
    s_vals = (C * Xs{ci})';
    plot(T{ci}, s_vals(:,1),'b','LineWidth',1.5,'DisplayName','s_1');
    plot(T{ci}, s_vals(:,2),'r','LineWidth',1.5,'DisplayName','s_2');
    yline(0,'Color',[0.6 0.6 0.6],'LineWidth',0.5,'LineStyle',':');
    title(labels{ci},'FontSize',10); xlabel('Time (s)'); ylabel('s');
    legend('Location','best','FontSize',9); grid on;
end

% Control Inputs
figure('Name','Q3 - Control Inputs','Position',[50 50 1200 700]);
ctrl_types = {'sf','sign','sat','sigmoid'};
eps_list = [0, 0, 0.01, 0.1];
for ci = 1:4
    subplot(2,2,ci); hold on;
    u1_arr = zeros(length(T{ci}),1);
    u2_arr = zeros(length(T{ci}),1);
    for k=1:length(T{ci})
        x = Xs{ci}(k,:);
        s = C*x;
        u = compute_u_q3(x,s,K_sf,K_eq,CB_inv,eta,ctrl_types{ci},eps_list(ci));
        u1_arr(k)=u(1); u2_arr(k)=u(2);
    end
    plot(T{ci}, u1_arr,'Color','red','LineWidth',1.2,'DisplayName','u_1');
    plot(T{ci}, u2_arr,'Color','blue','LineWidth',1.2,'DisplayName','u_2');
    yline(0,'Color',[0.6 0.6 0.6],'LineWidth',0.5,'LineStyle',':');
    title(labels{ci},'FontSize',10); xlabel('Time (s)'); ylabel('u');
    legend('Location','best','FontSize',9); grid on;
end

function dxdt = ode_q3(t, x, A, B, K_sf, K_eq, CB_inv, C, eta, d_func, type, eps)
    d = d_func(t);
    s = C * x;
    u = compute_u_q3(x, s, K_sf, K_eq, CB_inv, eta, type, eps);
    dxdt = A*x + B*(u + d);
end

```

```
function u = compute_u_q3(x, s, K_sf, K_eq, CB_inv, eta, type, eps)
    switch type
        case 'sf'
            u = -K_sf * x;
        case 'sign'
            u = -(K_eq*x) - CB_inv*(eta*sign(s));
        case 'sat'
            u = -(K_eq*x) - CB_inv*(eta*sat_vec(s,eps));
        case 'sigmoid'
            u = -(K_eq*x) - CB_inv*(eta*(s./(abs(s)+eps)));
    end
end

function v = sat_vec(s, eps)
    v = zeros(size(s));
    for i = 1:length(s)
        if s(i) > eps;          v(i) = 1;
        elseif s(i) < -eps;    v(i) = -1;
        else;                  v(i) = s(i)/eps;
    end
end
```

```
% Q. 2
```

```
clear; clc; close all;
```

```
m      = 4;
c      = 1;
eta    = 2;
k_func = @(t) 1.5 + 0.4*sin(2*t);
x0     = [2; 0.5];
tspan  = [0 10];
opts   = odeset('MaxStep', 0.001, 'RelTol', 1e-6);
eps_vals = [0.1, 0.5];
```

```
% 1. Ideal sign
```

```
[t_s, X_s] = ode45(@(t,x) dyn(t,x,m,c,eta,k_func,'sign',0), tspan, x0, opts);
```

```
% 2. Sigmoid and Saturation for each epsilon
```

```
results = struct();
results(1).label = 'sign(s)';          results(1).t = t_s; results(1).X = X_s; results(1).✓
type='sign'; results(1).eps=0;
idx = 2;
for eps = eps_vals
    [t_sig, X_sig] = ode45(@(t,x) dyn(t,x,m,c,eta,k_func,'sigmoid',eps), tspan, x0, ✓
opts);
    [t_sat, X_sat] = ode45(@(t,x) dyn(t,x,m,c,eta,k_func,'sat', eps), tspan, x0, opts);
    results(idx).label = sprintf('sigmoid \epsilon=%.1f',eps); results(idx).t=t_sig; ✓
results(idx).X=X_sig; results(idx).type='sigmoid'; results(idx).eps=eps; idx=idx+1;
    results(idx).label = sprintf('sat \epsilon=%.1f', eps); results(idx).t=t_sat; ✓
results(idx).X=X_sat; results(idx).type='sat';      results(idx).eps=eps; idx=idx+1;
end
```

```
% comparison at eps=0.1
```

```
cols = {'k','b','r'};
types = {'sign','sigmoid','sat'};
lbls = {'sign(s)', 'sigmoid \epsilon=0.1', 'sat \epsilon=0.1'};
```

```
figure('Name','Comparison eps=0.1','Position',[100 100 1100 750]);
```

```
vars = {'s(t)', 'u(t)', 'x_1(t)', 'x_2(t)'};
```

```
for vi = 1:4
    subplot(2,2,vi); hold on;
    for ri = [1 2 4] % sign, sigmoid_0.1, sat_0.1
        r = results(ri);
        s_arr = c*r.X(:,1) + r.X(:,2);
        u_arr = compute_u(r.t, r.X, m, c, eta, k_func, r.type, r.eps);
        data = {s_arr, u_arr, r.X(:,1), r.X(:,2)};
        ls = '--'; if ri>1; ls='-'; end
        plot(r.t, data{vi}, 'Color', cols(find([1 2 4]==ri)), 'LineWidth', ✓
1.5, 'LineStyle',ls, 'DisplayName',lbls(find([1 2 4]==ri)));
    end
    yline(0, 'Color',[0.6 0.6 0.6], 'LineWidth',0.5);
    xlabel('Time (s)'); ylabel(vars{vi}); title(vars{vi}); legend✓
('Location','best','FontSize',8); grid on;
```



```

end

% effect of epsilon on u(t)
figure('Name','epsilon effect','Position',[100 100 1200 500]);
ep_cols = {'b','g'};
for pi = 1:2
    subplot(1,2,pi);
    appr = {'sigmoid','sat'}; ttl = {'Sigmoid','Saturation'};
    % ideal sign
    u_sign = compute_u(t_s, X_s, m, c, eta, k_func, 'sign', 0);
    plot(t_s, u_sign, 'k--', 'LineWidth', 1.5, 'DisplayName', 'sign(s)'); hold on;
    for ei = 1:length(eps_vals)
        ri = 1 + (ei-1)*2 + (pi-1) + 1; % index into results
        r = results(ri);
        u_a = compute_u(r.t, r.X, m, c, eta, k_func, r.type, r.eps);
        plot(r.t, u_a, 'Color', ep_cols{ei}, 'LineWidth', 1.2, 'DisplayName', sprintf(
('\epsilon=%1f', r.eps)); hold on;
    end
    xlabel('Time (s)'); ylabel('u [N]'); title(ttl{pi}); legend('Location','best'); grid
on;
end

function dxdt = dyn(t, x, m, c, eta, k_func, type, eps)
    x1=x(1); x2=x(2); k=k_func(t);
    s = c*x1 + x2;
    switch type
        case 'sign',      sgn = sign(s);
        case 'sigmoid',  sgn = s/(abs(s)+eps);
        case 'sat',      sgn = sat_func(s,eps);
    end
    u = k*x2*abs(x2) - m*c*x2 - m*eta*sgn;
    dxdt = [x2; (u - k*x2*abs(x2))/m];
end

function u_arr = compute_u(t, X, m, c, eta, k_func, type, eps)
    u_arr = zeros(length(t),1);
    for i=1:length(t)
        x2=X(i,2); k=k_func(t(i)); s=c*X(i,1)+x2;
        switch type
            case 'sign',      sgn=sign(s);
            case 'sigmoid',  sgn=s/(abs(s)+eps);
            case 'sat',      sgn=sat_func(s,eps);
        end
        u_arr(i) = k*x2*abs(x2) - m*c*x2 - m*eta*sgn;
    end
end

function v = sat_func(s, eps)
    if s > eps;          v = 1;
    elseif s < -eps;    v = -1;
    else;                v = s/eps;
end

```

end
end

```

% Q. 4
clear; clc; close all;

f = @(x) x(1)^2 + 2*x(2);
g = @(x) 2;
d_func = @(t) 1.5 + 0.5*cos(t);

p = 5;
q = 3;
beta = 1;
eta = 5;
d_nom = 1.5;

function dxdt = ntsmc_ode(t, x, f_func, g_func, d_func, p, q, beta, eta, d_nom)
    x1 = x(1); x2 = x(2);
    d = d_func(t);
    fx = f_func(x);
    gx = g_func(x);

    s = x1 + (1/beta) * sign(x2) * abs(x2)^(p/q);
    coeff = (beta * q) / (p); % coefficient from surface derivative
    u = -(1/gx) * (fx + coeff * abs(x2)^(2 - p/q) + eta*sign(s)) - d_nom;

    dxdt = [x2;
            fx + gx*(u + d)];
end

x0 = [2; 3.5];
tspan = [0 10];
opts = odeset('MaxStep',0.001,'RelTol',1e-7,'AbsTol',1e-8);

[t, X] = ode45(@(t,x) ntsmc_ode(t,x,f,g,d_func,p,q,beta,eta,d_nom), tspan, x0, opts);

s_arr = X(:,1) + (1/beta)*sign(X(:,2)).*abs(X(:,2)).^(p/q);
u_arr = zeros(length(t),1);
d_arr = zeros(length(t),1);
for i = 1:length(t)
    xi = X(i,:);
    fx = f(xi); gx = g(xi);
    di = d_func(t(i));
    d_arr(i) = di;
    si = s_arr(i);
    coeff = (beta*q)/p;
    u_arr(i) = -(1/gx)*(fx + coeff*abs(xi(2))^(2-p/q) + eta*sign(si)) - d_nom;
end

% State trajectories and sliding variable
figure('Name','NTSMC','Position',[100 100 1100 800]);

subplot(2,2,1)
plot(t, X(:,1),'b','LineWidth',2); hold on;

```

```
ylabel(0,'k--','LineWidth',0.8);
xlabel('Time (s)'); ylabel('x_1 [m]'); title('State x_1(t)'); grid on;

subplot(2,2,2)
plot(t, X(:,2),'m','LineWidth',2); hold on;
ylabel(0,'k--','LineWidth',0.8);
xlabel('Time (s)'); ylabel('x_2 [m/s]'); title('State x_2(t)'); grid on;

subplot(2,2,3)
plot(t, s_arr,'r','LineWidth',1.8); hold on;
ylabel(0,'k--','LineWidth',0.8);
xlabel('Time (s)'); ylabel('s'); title('Sliding Variable s(t)'); grid on;

subplot(2,2,4)
plot(t, u_arr,'Color',[0.1 0.6 0.1],'LineWidth',1.5);
xlabel('Time (s)'); ylabel('u'); title('Control Input u(t)'); grid on;

% Phase portrait x2 vs x1
figure('Name','Phase Portrait','Position',[200 200 600 500]);
plot(X(:,1), X(:,2),'b','LineWidth',2); hold on;
plot(X(1,1), X(1,2), 'go','MarkerSize',10,'MarkerFaceColor','g','DisplayName','Start');
plot(X(end,1),X(end,2), 'rs','MarkerSize',10,'MarkerFaceColor','r','DisplayName','End');

% Plot sliding manifold (s=0 curve):  $x_1 = -(1/\beta) \cdot \text{sign}(x_2) \cdot |x_2|^{(p/q)}$ 
x2_range = linspace(-4, 4, 400);
x1_manifold = -(1/beta)*sign(x2_range).*abs(x2_range).^(p/q);
plot(x1_manifold, x2_range,'r--','LineWidth',1.5,'DisplayName','Sliding manifold s=0');

xlabel('x_1'); ylabel('x_2'); title('Phase Portrait');
legend('Trajectory','Start','End','s=0 manifold','Location','best');
grid on;
```